

SKILLS

Programming & Tools: Python (NumPy, pandas, SciPy, scikit-learn, PyTorch, Matplotlib) — primary language, 5+ years; SQL; IDL; Bash, Git, $\LaTeX$; Excel and Microsoft Office; Linux/UNIX and HPC environments.

Data & ML: Statistical analysis, data cleaning and validation, feature engineering, supervised classification with neural networks, PCA, time-series analysis, data visualisation.

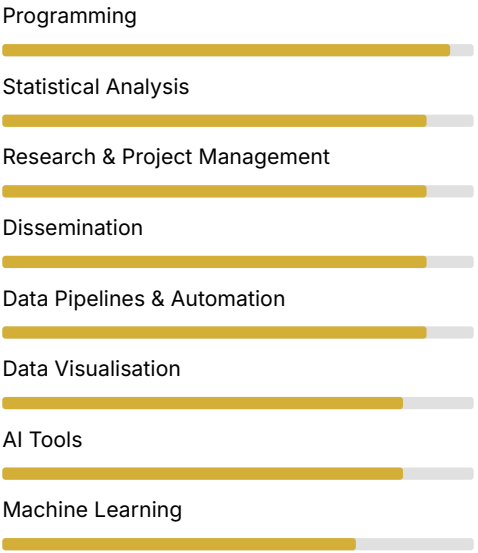
Data Management: Multi-terabyte datasets; remote compute and server access; database access; pipeline and data-efficiency management; data reduction.

Data Coordination: Multi-source data alignment; data calibration; operations management.

AI Tools: Proficient with a range of AI models; experienced with Claude Code and Cowork; usage management, token efficiency, and quality control.

Communication: Technical writing — peer-reviewed publications and code documentation; referee for scientific journals; reporting results to senior advisors and project leaders; conference talks, invited seminars, and public outreach.

Leadership: Supervised a research intern and MSci students; chaired a session at an international scientific meeting; managed an amateur football team, including finances, sponsorship, and competition entries.



PROFILE

Research scientist working in applied computational astrophysics seeking to transition from an academic environment into a fast-paced, high-skilled, technical role. I have over five years of experience in independent and collaborative quantitative research, involving project conception, funding acquisition, data analysis, results delivery, and defence of findings. My work is focussed on the analysis of large, multi-source satellite datasets, statistical modelling, and machine-learning classification, with results published in peer-reviewed journals and presented across a spectrum of audiences, from NASA mission teams to international conferences to public outreach talks. I am now looking to transition into an industry where I can utilise my analytical skill set to provide science-quality solutions to real-world commercial problems.

EXPERIENCE

Postdoctoral Research Fellow

2025–Present

Astrophysics Research Centre, Queen’s University Belfast

- Led the largest statistical study to date in my field, analysing 15 years of NASA satellite data covering several thousand events. I designed and ran the full Python pipeline — ingestion, cleaning, feature extraction, and classification — and published the results as first author in June 2026.
- Built supervised machine-learning models in PyTorch to detect weak signals in noisy time-series data.
- Ran GPU-enabled workloads in high-performance and remote computing environments, managing compute-time allocation and pipeline efficiency.
- Work with global mission development teams across the US, Europe, and Japan.
- Conceived and delivered a team research project from conceptualisation through to publication: scoped the project, recruited and supervised a research intern, and managed the work to a peer-reviewed result; have mentored MSci students since 2022.
- Presented research to international audiences at conferences, invited seminars, and public outreach talks.

Visiting Scientist

Jan–May 2023

NASA Goddard Space Flight Center, Maryland, USA

- Secured competitive funding for overseas fieldwork, conducting a collaborative research project with mission scientists on the NASA Goddard campus.

Sales Assistant

2015–2020

Millets, Derby

- Met individual and team sales targets over five years in a KPI-driven store, working part-time alongside full-time study.

EDUCATION

PhD in Astrophysics

2021–2025

Queen’s University Belfast

Thesis: *Observational Analysis of Lyman-alpha Emission from Solar Flares* — an extensive multi-instrument, multi-mission analysis of the solar plasma environment, particle properties, and terrestrial impact of the Sun’s most powerful emission.

MSci in Physics (First Class Honours)

2017–2021

Queen’s University Belfast

Thesis: *Computational Modelling of the Chromospheric Response to Solar Flares* — used 1D and 3D models of solar plasma properties to simulate particle interactions following a solar flare.

RESEARCH HIGHLIGHTS

The EUV Late-Phase: Statistical Results from 15 Years of Solar Dynamics Observatory Observations

Greatorex, H. J., et al. — *Solar Physics*, 301, 6 (2026)

DOI: 10.1007/s11207-026-02673-7

On the Instrumental Discrepancies in Lyman-alpha Observations of Solar Flares

Greatorex, H. J., et al. — *Solar Physics*, 299, 162 (2024)

DOI: 10.1007/s11207-024-02407-7

Observational Analysis of Lyman-alpha Emission in Equivalent Magnitude Solar Flares

Greatorex, H. J., et al. — *The Astrophysical Journal*, 954, 120 (2023)

DOI: 10.3847/1538-4357/acea7f

REFERENCES

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Line manager and former PhD supervisor

Prof. Dr Louise Harra

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Collaborator